EDULAB NIGERIA

PRACTICAL DIGITAL SKILLS CURRICULUM

Primary 3 – Senior Secondary 2 | 2026/2027

Nine self-contained annual courses • 24 weeks per class • Three applied projects per class

A grant-submission curriculum framework and implementation plan

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Document status: proposed implementation curriculum. Alignment to particular national examinations, standards or accreditation requirements should be independently mapped and verified before any compliance claim.

1. Programme overview

EduLab Nigeria partners with schools to provide practical computer education through school-based labs, trained instructors, structured lessons and project assessment. This curriculum is designed for learners from Primary 3 through SS2. Every class is an independent 24-week annual course: a new student can join at the beginning of any class without having attended a lower EduLab class.

Design element	Commitment
Class levels	Primary 3, Primary 4, Primary 5, Primary 6, JSS1, JSS2, JSS3, SS1 and SS2
Annual contact time	24 instructional weeks, 2 hours weekly = 48 guided hours
Term length	8 weeks × 3 terms; 16 guided hours each term
Project requirement	One practical assessed term project; 3 projects per class annually
Equipment	School ICT laboratory; initial implementation assumes 10 PCs per partner school
Suggested practice model	Pairs or rotating small groups as needed; learners assessed individually
Delivery	Offline-first where possible; supervised internet when lesson requires it
Entry rule	No prerequisite from earlier EduLab classes; term-one orientation and recap embedded

2. Curriculum design and safeguards

• Each class starts with an orientation to required devices, file management, safe computer use and the basics of that class's selected software. No teacher should assume students completed a preceding EduLab class.

• Within a class, the three terms are designed as a progression. Each new term nevertheless includes a short starter/refresher or ready-made starter assets, so late entrants can participate.

• Teach hands-on: instructor demonstration, guided practice, individual application, peer feedback and saving work. Use local examples relevant to Nigeria.

• Projects use anonymised or fictional pupil data. Never publish student names, contact details or photographs without required permission.

• For minors, avoid unsupervised creation of external public accounts. Use offline applications, teacher-managed tools or age-appropriate institutional accounts.

• Where internet is unavailable, use offline installers, local documents/datasets and offline Scratch. CBT practice software is installed locally where supported.

• Adapt tasks for learners with disabilities; offer keyboard alternatives, high contrast, larger text and flexible response formats.

3. Teaching and assessment standard

Component	Suggested approach
Weekly lesson (120 minutes)	10 min retrieval/starter; 20 min concept and demo; 65 min guided/independent lab; 15 min feedback/debugging; 10 min save, reflection and exit ticket
Evidence	Save editable source, final output/screenshot, short reflection and teacher checklist
Term evaluation	Hands-on weekly skills (30%); final project (50%); demonstration/explanation (20%)

Minimum project standard	Works as intended; shows essential term skills; complete files; clear explanation and safe/ethical practice
Individual verification	Each learner demonstrates at least one key action or explains an essential design choice
Term 8	Final build, testing, showcase, assessment and portfolio archive
Progress monitoring	Pre-lesson skills check, weekly skills log, term projects and end-of-year learner portfolio

4. Class overview and assessed projects

Class	Independent annual course	Term 1 project	Term 2 project	Term 3 project
Primary 3	Computer Confidence, Typing and Digital Storytelling	My Computer Skills Portfolio	My School Routine	My Digital Storybook
Primary 4	Digital Research and Publishing	Classroom Rules Poster	Animal Fact Sheet	My Community Newsletter
Primary 5	Creative Coding with Scratch	Interactive Animated Story	Interactive Quiz	Catch-the-Star Game
Primary 6	Computational Thinking and Scratch Projects	Animated Science Explainer	Math Practice Game	Community Learning App
JSS 1	Front-End Web Development — Foundations	My School Information Site	Styled School Website	Interactive Student Portal
JSS 2	Python Programming and Problem Solving	Student Study Planner	Student Result Calculator	School Library Inventory
JSS 3	Data Literacy, Spreadsheets and Databases	Class Results Workbook	School Survey Dashboard	School Club Member Register
SS 1	Python Applications and Web Foundations	Personal Study Assistant	School Notice Board	WAEC/CBT Readiness Portfolio
SS 2	Applied Data, Responsible AI and Digital Entrepreneurship	Nigeria Data Story	AI-Assisted Study Resource	Community Tech Solution Pitch

Projects and lesson sequences are proposals for deployment and grant review, not claims of outcomes already achieved.

1. Primary 3 — Computer Confidence, Typing and Digital Storytelling

Entry: Open to new learners at this class level. No prior computer experience required. Learners gain confidence using a PC, produce a document, and tell a story digitally.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Computer Foundations

Term outcome: Identify and operate the main parts of a computer safely.

Week	Focus	Practical learning activity / observable evidence
01	Meet the computer	Identify monitor, system unit, keyboard, mouse and headphones.
02	Lab rules and care	Practise safe handling, posture, sharing and basic safety.
03	Switch on and shut down	Follow the correct startup, login and shutdown sequence.
04	Mouse control	Point, click, double-click, drag and drop in a practice activity.
05	Keyboard basics	Type letters, numbers, space, enter and backspace.
06	Files and folders	Create a named folder and save a teacher-provided file.
07	Prepare portfolio	Combine an illustrated diagram and a skills checklist.
08	Term project and showcase	Complete, test, present and submit “My Computer Skills Portfolio”; evidence: Computer-parts diagram, safe-start checklist, saved practice file.

Assessed term project: My Computer Skills Portfolio. Computer-parts diagram, safe-start checklist, saved practice file.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Typing and School Documents

Term outcome: Type, format and save a simple school document.

Week	Focus	Practical learning activity / observable evidence
09	Open a word processor	Start LibreOffice Writer or equivalent and type a name.
10	Typing sentences	Enter short sentences with capitals and punctuation.
11	Edit text	Select, delete, correct and move text.
12	Format text	Change title size, bold, colour and alignment.
13	Lists and spacing	Create a clear list of school activities.
14	Insert simple table	Build a two-column daily timetable.
15	Save and export	Save an editable copy and create a PDF.
16	Term project and showcase	Complete, test, present and submit “My School Routine”; evidence: One-page document with title, list, timetable and PDF.

Assessed term project: My School Routine. One-page document with title, list, timetable and PDF.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Digital Art and Stories

Term outcome: Make original digital illustrations and a short story.

Week	Focus	Practical learning activity / observable evidence
17	Open painting software	Find brushes, lines, colours and erase tools.
18	Draw shapes	Draw a school, tree or home using basic shapes.
19	Plan a story	Identify character, setting and beginning/middle/end.
20	Make character images	Draw and save two original character pictures.
21	Combine pictures and text	Put one image and short text on each page.
22	Improve readability	Check spelling, alignment, font size and contrast.
23	Export storybook	Save source files and export the booklet to PDF.
24	Term project and showcase	Complete, test, present and submit “My Digital Storybook”; evidence: Three-page illustrated story with opening, middle and ending.

Assessed term project: My Digital Storybook. Three-page illustrated story with opening, middle and ending.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

2. Primary 4 — Digital Research and Publishing

Entry: Open to new learners at this class level. Begins with essential computer and typing practice; no Primary 3 completion required.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Creating Useful Posters

Term outcome: Use drawing and page-layout tools to communicate a clear message.

Week	Focus	Practical learning activity / observable evidence
01	Starter computer skills	Practise mouse, keyboard, open/save and lab rules.
02	Purpose and audience	Identify what makes a poster clear and useful.
03	Create headings	Type a short title and subheadings in Writer.
04	Page layout	Use margins, spacing, alignment and readable fonts.
05	Pictures and shapes	Insert or draw original icons and pictures.
06	Colour and hierarchy	Use contrast and visual order responsibly.
07	Revise and export	Proofread, peer-review, save PDF.
08	Term project and showcase	Complete, test, present and submit "Classroom Rules Poster"; evidence: A4 poster, planning sketch and final PDF.

Assessed term project: Classroom Rules Poster. A4 poster, planning sketch and final PDF.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Research and Fact Sheets

Term outcome: Research safely using curated offline/online sources and write factual text.

Week	Focus	Practical learning activity / observable evidence
09	Computer review and safety	Open files, rename and practise safe browsing.
10	Research questions	Choose one animal and draft four questions.
11	Finding reliable facts	Use teacher-curated Kiwix/offline encyclopaedia or safe sites.
12	Notes in own words	Record short facts without copying long paragraphs.
13	Tables and labels	Create a habitat/food/lifespan table.
14	Images and credits	Insert permitted image and name its source.
15	Draft and review	Build and edit a two-page fact sheet.
16	Term project and showcase	Complete, test, present and submit "Animal Fact Sheet"; evidence: Two-page fact sheet with image, table, sources and PDF.

Assessed term project: Animal Fact Sheet. Two-page fact sheet with image, table, sources and PDF.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Community Publishing

Term outcome: Plan and publish a simple evidence-based newsletter.

Week	Focus	Practical learning activity / observable evidence
17	What is a newsletter?	Examine headings, articles, captions and sections.
18	Interview safely	Prepare questions and interview a classmate or teacher.
19	Report local information	Write a short school/community story.
20	Photo and caption	Use permitted photographs or drawings with captions.
21	Create page layout	Place titles, text blocks and pictures.
22	Check facts and consent	Verify names and do not publish private information.
23	Finish publication	Edit, combine stories and export final PDF.
24	Term project and showcase	Complete, test, present and submit “My Community Newsletter”; evidence: Two-page illustrated school/community newsletter and PDF.

Assessed term project: My Community Newsletter. Two-page illustrated school/community newsletter and PDF.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

3. Primary 5 — Creative Coding with Scratch

Entry: Open to new learners at this class level. A fresh introduction to visual programming; no prior coding needed.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Scratch Basics and Storytelling

Term outcome: Understand sequences, sprites, events and movement.

Week	Focus	Practical learning activity / observable evidence
01	Digital safety and Scratch	Open Scratch offline/editor and save a starter project.
02	Algorithm and sequence	Write step-by-step instructions for a daily activity.
03	Stage and sprites	Choose, draw and move characters.
04	Events and motion	Use when-green-flag and motion blocks.
05	Looks and speech	Create dialogue, costumes and backdrops.
06	Timing and sound	Use waits, sequences and permitted sound.
07	Build and debug	Combine scenes and test story flow.
08	Term project and showcase	Complete, test, present and submit "Interactive Animated Story"; evidence: Scratch animation with 2 scenes, 2 characters and narration/text.

Assessed term project: Interactive Animated Story. Scratch animation with 2 scenes, 2 characters and narration/text.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Logic and Interaction

Term outcome: Use conditions, sensing and repetition to build interactive behaviour.

Week	Focus	Practical learning activity / observable evidence
09	Review in this term	Create a small Scratch script from scratch.
10	Loops	Repeat actions with repeat and forever blocks.
11	Conditions	Make if/else decisions using simple examples.
12	Sensing and input	Ask questions and receive typed answers.
13	Variables	Create a score and reset it.
14	Feedback	Display correct/incorrect responses.
15	Test with classmates	Fix scoring, question flow and restart logic.
16	Term project and showcase	Complete, test, present and submit "Interactive Quiz"; evidence: Scratch quiz with at least 5 questions, feedback and restart.

Assessed term project: Interactive Quiz. Scratch quiz with at least 5 questions, feedback and restart.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Game Design

Term outcome: Combine events, logic and variables in a playable game.

Week	Focus	Practical learning activity / observable evidence
17	Mini-game introduction	Build a simple moving sprite and save.
18	Player controls	Use arrow keys or mouse to move a character.
19	Objects and randomness	Position objects using random coordinates.
20	Collision detection	Detect touching and respond.
21	Score and time	Use variables for score or countdown.
22	Rules and levels	Create goal, instructions and end state.
23	Debug and polish	Peer-test, improve difficulty and document controls.
24	Term project and showcase	Complete, test, present and submit "Catch-the-Star Game"; evidence: Playable game with scoring, instructions and win/lose condition.

Assessed term project: Catch-the-Star Game. Playable game with scoring, instructions and win/lose condition.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

4. Primary 6 — Computational Thinking and Scratch Projects

Entry: Open to new learners at this class level. Begins with a Scratch introduction and builds toward independent educational software.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Scratch Starter to Designer

Term outcome: Learn visual-programming foundations and build animated lessons.

Week	Focus	Practical learning activity / observable evidence
01	Scratch from zero	Learn editor, sprites, stage, save/open.
02	Algorithms	Turn a simple explanation into ordered steps.
03	Events and sequences	Start scenes using events and messages.
04	Loops and timing	Repeat actions with timing.
05	Costumes and visuals	Create labelled illustrations.
06	Sound and narration	Add appropriate audio/text explanations.
07	Prototype and test	Review factual clarity and timing.
08	Term project and showcase	Complete, test, present and submit "Animated Science Explainer"; evidence: Two-minute Scratch explanation of a science or health concept.

Assessed term project: Animated Science Explainer. Two-minute Scratch explanation of a science or health concept.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Smart Interactive Games

Term outcome: Create a game using conditionals, variables and feedback.

Week	Focus	Practical learning activity / observable evidence
09	Warm-up blocks	Rebuild a basic event and movement script.
10	User input	Ask for answers using sensing.
11	Operators	Compare answers with numbers and strings.
12	Conditions	Choose feedback based on correctness.
13	Variables	Track score and questions completed.
14	Broadcasts	Organise different quiz scenes.
15	Debug and usability	Test wrong answers and reset behaviour.
16	Term project and showcase	Complete, test, present and submit "Math Practice Game"; evidence: 10-question arithmetic game with scoring, replay and instructions.

Assessed term project: Math Practice Game. 10-question arithmetic game with scoring, replay and instructions.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Educational App Design

Term outcome: Plan, prototype and evaluate an educational Scratch application.

Week	Focus	Practical learning activity / observable evidence
17	Define learning problem	Pick a local topic and target user.
18	Plan user journey	Sketch start, instructions, content and result screens.
19	Create assets	Design sprites/backdrops with readable text.
20	Interactive lesson	Build content with events and sequences.
21	Knowledge check	Add scoring/feedback and safe interactions.
22	Accessibility basics	Improve fonts, keyboard access and contrast.
23	User testing	Collect three testers' feedback and refine.
24	Term project and showcase	Complete, test, present and submit "Community Learning App"; evidence: Scratch application teaching a chosen topic with evaluation checklist.

Assessed term project: Community Learning App. Scratch application teaching a chosen topic with evaluation checklist.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

5. JSS 1 — Front-End Web Development — Foundations

Entry: Open to new learners at this class level. New entrants begin with no coding assumed. Term 1 HTML, Term 2 CSS, Term 3 JavaScript.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: HTML — Structure

Term outcome: Create semantic multi-page HTML documents.

Week	Focus	Practical learning activity / observable evidence
01	How websites work	Explain browser, website, file and editor; create project folder.
02	First HTML document	Write doctype, html, head, title and body.
03	Text and headings	Use headings, paragraphs, strong and emphasis.
04	Lists and links	Create ordered/unordered lists and internal links.
05	Images and accessibility	Insert local images with alt descriptions.
06	Semantic structure	Use header, nav, main, section and footer.
07	Forms basics and validation	Add labelled inputs; test links and HTML structure.
08	Term project and showcase	Complete, test, present and submit "My School Information Site"; evidence: Three linked HTML pages with headings, images, lists and navigation.

Assessed term project: My School Information Site. Three linked HTML pages with headings, images, lists and navigation.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: CSS — Visual Design

Term outcome: Style and lay out an accessible web page.

Week	Focus	Practical learning activity / observable evidence
09	CSS from zero	Style a supplied HTML starter; selectors and rules.
10	Colour and typography	Choose readable font, background and contrast.
11	Box model	Practise margin, padding and borders.
12	Layout with Flexbox	Build a navigation and two-column cards.
13	Images and spacing	Style responsive images and sections.
14	Responsive design	Use media queries for smaller screens.
15	Review design	Fix accessibility, alignment and cross-browser issues.
16	Term project and showcase	Complete, test, present and submit "Styled School Website"; evidence: Responsive version of the Term 1 or supplied starter site.

Assessed term project: Styled School Website. Responsive version of the Term 1 or supplied starter site.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: JavaScript — Interactivity

Term outcome: Add basic events and logic to a front-end site.

Week	Focus	Practical learning activity / observable evidence
17	JavaScript introduction	Use script tag, console and variables.
18	Data and operators	Work with strings, numbers and comparisons.
19	Conditions	Display feedback using if/else.
20	Functions	Write reusable functions for user actions.
21	DOM basics	Select and update text and HTML elements.
22	Events and forms	Handle clicks and check form input.
23	Integrate and test	Add interactive quiz or greeting to a starter site.
24	Term project and showcase	Complete, test, present and submit "Interactive Student Portal"; evidence: Front-end mock student portal with form validation and simple quiz.

Assessed term project: Interactive Student Portal. Front-end mock student portal with form validation and simple quiz.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

6. JSS 2 — Python Programming and Problem Solving

Entry: Open to new learners at this class level. Starts programming at beginner level; does not rely on JSS1 web development.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Python Fundamentals

Term outcome: Write small programs using variables, input and decisions.

Week	Focus	Practical learning activity / observable evidence
01	Getting started with Python	Run Python, print text and save a script.
02	Variables and types	Use strings, integers and simple conversions.
03	Input and output	Ask user for name, class and subjects.
04	Arithmetic	Calculate sums, time totals and percentages.
05	Conditions	Make decisions with if/elif/else.
06	Loops	Use for and while loops safely.
07	Functions and debugging	Break program into functions and test sample inputs.
08	Term project and showcase	Complete, test, present and submit "Student Study Planner"; evidence: Python console app asking subjects and displaying a study plan.

Assessed term project: Student Study Planner. Python console app asking subjects and displaying a study plan.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Collections and Data

Term outcome: Organise data with lists and dictionaries.

Week	Focus	Practical learning activity / observable evidence
09	Python refresher	Revisit variables, loops, input and functions.
10	Lists	Store subject names and scores.
11	List operations	Append, remove, iterate and sort.
12	Dictionaries	Create a student record using keys and values.
13	Nested data	Store several subjects for one student.
14	Calculations and grading	Calculate mean and grade boundaries.
15	Error handling and testing	Handle missing or invalid numeric inputs.
16	Term project and showcase	Complete, test, present and submit "Student Result Calculator"; evidence: Python program that stores scores and calculates totals, averages and grades.

Assessed term project: Student Result Calculator. Python program that stores scores and calculates totals, averages and grades.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Python Project Delivery

Term outcome: Plan, build, test and explain a complete program.

Week	Focus	Practical learning activity / observable evidence
17	Program brief	Identify inputs, outputs and sample users.
18	Modules and files	Read/write simple CSV or text records.
19	Data design	Plan book title, ID and availability fields.
20	Functions and menu	Add, find, borrow and return actions.
21	Validation	Prevent blank entries and duplicate IDs.
22	Testing	Prepare and run at least 8 test cases.
23	Documentation	Write usage guide and demonstrate.
24	Term project and showcase	Complete, test, present and submit “School Library Inventory”; evidence: Console-based lending register with search and file save.

Assessed term project: School Library Inventory. Console-based lending register with search and file save.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

7. JSS 3 — Data Literacy, Spreadsheets and Databases

Entry: Open to new learners at this class level. Self-contained data course; no prior Python or spreadsheets expected.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Spreadsheets for Decisions

Term outcome: Enter, clean and calculate classroom data.

Week	Focus	Practical learning activity / observable evidence
01	Data and spreadsheet basics	Open spreadsheet, cells, rows, columns, save.
02	Data entry	Create a small fictional result sheet.
03	Formulas	Use +, -, SUM and AVERAGE.
04	References	Apply formulas using relative references.
05	Sorting and filtering	Locate highest and lowest results.
06	Charts	Create a bar chart from aggregated totals.
07	Check errors	Verify formulas and protect private student data.
08	Term project and showcase	Complete, test, present and submit "Class Results Workbook"; evidence: Workbook with scores, totals, averages and grade summary.

Assessed term project: Class Results Workbook. Workbook with scores, totals, averages and grade summary.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Data Visualisation and Reporting

Term outcome: Turn a small dataset into useful, honest charts.

Week	Focus	Practical learning activity / observable evidence
09	Refresh spreadsheet basics	Use a starter dataset and cell formulas.
10	Collect and define data	Create safe multiple-choice survey questions.
11	Data quality	Find blanks, duplicate entries and inconsistencies.
12	Counts and categories	Use COUNTIF and simple pivot-style summaries.
13	Charts for purpose	Select column/bar or pie chart appropriately.
14	Interpret results	Write three cautious insights.
15	Create dashboard	Organise charts, sources and explanatory captions.
16	Term project and showcase	Complete, test, present and submit "School Survey Dashboard"; evidence: Survey spreadsheet with 20+ fictional responses, two charts and findings.

Assessed term project: School Survey Dashboard. Survey spreadsheet with 20+ fictional responses, two charts and findings.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Database Fundamentals

Term outcome: Use tables, fields and forms to organise information.

Week	Focus	Practical learning activity / observable evidence
17	Database concepts	Understand records, fields, tables and keys.
18	Plan safe dataset	Use invented student names and IDs.
19	Create tables	Define appropriate field types.
20	Data entry forms	Add records using a simple form.
21	Queries and filters	Find members by club or level.
22	Reports	Create a simple membership report.
23	Data protection and testing	Check duplicates, access and backup.
24	Term project and showcase	Complete, test, present and submit "School Club Member Register"; evidence: LibreOffice Base / Access or spreadsheet-backed database with queries.

Assessed term project: School Club Member Register. LibreOffice Base / Access or spreadsheet-backed database with queries.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

8. SS 1 — Python Applications and Web Foundations

Entry: Open to new learners at this class level. Begins coding fundamentals within the class and moves from scripts to local web applications.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Python from Scratch

Term outcome: Program basic automation and use files safely.

Week	Focus	Practical learning activity / observable evidence
01	Python beginner setup	Run scripts, print and use editor.
02	Variables and input	Work with strings, numbers and conversions.
03	Conditions	Implement simple decision rules.
04	Loops	Repeat tasks and menu selections.
05	Functions	Separate tasks into named functions.
06	Working with files	Store simple records in text/CSV.
07	Test and document	Create usage steps and verify edge cases.
08	Term project and showcase	Complete, test, present and submit “Personal Study Assistant”; evidence: Console application with subject plan, reminders and saved data.

Assessed term project: Personal Study Assistant. Console application with subject plan, reminders and saved data.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: Web Applications with Flask

Term outcome: Build and run a local web app with HTML templates.

Week	Focus	Practical learning activity / observable evidence
09	Web fundamentals from zero	Define browser, URL, HTML, server and route.
10	Flask setup	Create a local environment and Hello World route.
11	Templates	Create base HTML and template inheritance.
12	Forms	Send safe input with GET/POST.
13	SQLite basics	Create and retrieve notices.
14	Basic validation	Reject blank content and escape output.
15	Testing and demo	Run app on localhost, capture screenshots and describe limits.
16	Term project and showcase	Complete, test, present and submit “School Notice Board”; evidence: Local Flask app with pages and simple SQLite-backed notices.

Assessed term project: School Notice Board. Local Flask app with pages and simple SQLite-backed notices.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Applied Technology and CBT

Term outcome: Use digital productivity, assessment and documentation tools.

Week	Focus	Practical learning activity / observable evidence
17	Digital foundations	Revisit file management, backups and safety.
18	Spreadsheet analysis	Use SUM, AVERAGE, charts and clean entries.
19	CBT interface	Practise navigation, timer, review and submit.
20	Exam strategy	Review errors using topic-based feedback.
21	Documentation	Write README and annotate screenshots.
22	Portfolio packaging	Organise projects with attribution.
23	Presentation and improvement	Present progress with assessment evidence.
24	Term project and showcase	Complete, test, present and submit "WAEC/CBT Readiness Portfolio"; evidence: Spreadsheet, timed practice reflection and 2 documented coding samples.

Assessed term project: WAEC/CBT Readiness Portfolio. Spreadsheet, timed practice reflection and 2 documented coding samples.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

9. SS 2 — Applied Data, Responsible AI and Digital Entrepreneurship

Entry: Open to new learners at this class level. Begins with tool orientation; neither SS1 Python nor previous EduLab classes are required.

Annual outcomes: learners complete three term projects, store their work in a class portfolio and explain the safety and practical skills used.

Term 1: Data Analysis from First Principles

Term outcome: Clean, summarise and communicate simple data.

Week	Focus	Practical learning activity / observable evidence
01	Data questions	Choose an education or community question.
02	Data tables	Open CSV data in spreadsheet; learn rows, fields and types.
03	Cleaning	Fix missing values, formats and duplicates.
04	Statistics	Calculate sums, averages, percentages.
05	Charts	Choose and produce suitable visuals.
06	Interpretation	Distinguish patterns from causal claims.
07	Storytelling	Build concise report with chart descriptions and source.
08	Term project and showcase	Complete, test, present and submit "Nigeria Data Story"; evidence: Data report with dataset source, charts and three findings.

Assessed term project: Nigeria Data Story. Data report with dataset source, charts and three findings.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 2: AI Literacy and Responsible Use

Term outcome: Understand AI capabilities, limits and ethical use through accessible tools.

Week	Focus	Practical learning activity / observable evidence
09	What AI can and cannot do	Discuss pattern recognition and examples in Nigeria.
10	Data and models	Explain training data, examples and bias simply.
11	Prompts and evaluation	Write precise questions; compare outputs.
12	Verification	Check generated facts against trusted sources.
13	Safety and privacy	Avoid personal/sensitive data; discuss plagiarism.
14	AI in education	Explore permitted offline or supervised AI examples.
15	Prepare guided project	Document input, output, checks and human contribution.
16	Term project and showcase	Complete, test, present and submit "AI-Assisted Study Resource"; evidence: Fact-checked learning guide with prompts, evidence and AI-use

	disclosure.

Assessed term project: AI-Assisted Study Resource. Fact-checked learning guide with prompts, evidence and AI-use disclosure.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Term 3: Digital Solutions and Entrepreneurship

Term outcome: Develop and pitch a small socially useful technology concept.

Week	Focus	Practical learning activity / observable evidence
17	Problem discovery	Interview safely and select local problem.
18	User research	Record three anonymised needs.
19	Solution planning	Define features and project constraints.
20	Prototype	Make a simple site, dashboard or clickable mockup.
21	Cost and pricing	Estimate costs and sources of sustainability.
22	Testing and iteration	Collect feedback; record improvements.
23	Pitch preparation	Prepare demo, impact story and clear next steps.
24	Term project and showcase	Complete, test, present and submit "Community Tech Solution Pitch"; evidence: Prototype/mockup, one-page business model, budget and pitch presentation.

Assessed term project: Community Tech Solution Pitch. Prototype/mockup, one-page business model, budget and pitch presentation.

Assessment checks: functional or complete output; accurate key skills; file organisation; explanation; safety and originality.

Annual portfolio

Required: three finished projects (editable files where available), three final exports/screenshots, a weekly skills checklist, and a short learner reflection on what they can now do independently.

5. Equipment, software and implementation requirements

Area	Recommended provision
Lab hardware	10 functioning computer systems per school initially, projector where funded, tables, appropriate seating and reliable school electricity
Core offline software	LibreOffice Writer/Calc/Impress, Scratch offline, painting application, browser, Python 3, suitable code editor and locally deployed EduLab CBT platform
Additional software	Flask and SQLite for SS1; safe data tools or browser AI demonstrations for SS2; use age-appropriate alternatives as available
Account policy	Teacher-controlled or offline accounts for minors; obtain school/guardian consent when an online service is necessary
Instructor plan	One instructor may serve two nearby schools only with a written timetable that meets 2 hours per class weekly and provides time for maintenance and assessment
Learner scheduling	Rotate classes and small groups based on 10-PC lab capacity;

	log individual hands-on access
Recordkeeping	Weekly attendance, skills checklist, equipment incident log, term project rubric and annual report

6. Implementation milestones

1. Before launch: baseline learner skills check, schedule, equipment testing, safe-use induction and teacher preparation.

2. Weeks 1–8: core term skills taught; first projects submitted, scored and exhibited.

3. Weeks 9–16: new term starter, projects built and feedback given; mid-year progress review.

4. Weeks 17–24: final term course and projects; learner showcase, portfolio moderation and parent/school report.

5. After session: review project completion, attendance, device uptime, student feedback and school satisfaction; revise curriculum as needed.

7. Expected outcomes and monitoring

Monitoring indicators will be reported as actual counts and percentages after deployment, not assumed achievements: enrolled learners; attendance; proportion reaching minimum practical skill benchmark; term project completion; CBT navigation confidence; instructor contact hours; working device availability; student/parent/school satisfaction; and cost per learner.

8. Source and design notes

Developed with reference to EduLab Nigeria's previously supplied full curriculum (2026), its detailed Primary 3 and Primary 4 practical manuals (2026/27), and founder requirements for nine standalone class-level programmes and one project each term. The previous full curriculum grouped some primary classes together and included prerequisite-dependent topics in some secondary classes; the present proposal deliberately separates Primary 3–6 and provides fresh entry points for every class.

Edits for this grant version: excluded Primary 1–2 and SS3, consistent with the current target range; reserved 8 weeks per term; added individual outcomes, assessable evidence, offline delivery and safeguarding; adjusted class topics to avoid prerequisites from an earlier EduLab class. These are design decisions, not independently verified government curriculum alignments.

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